

Engine C — The Digital Recyclability Passport: Maximum-Ceiling Technical & Regulatory Specification for REWEAR-FUSED

TL;DR

- **The maximum-ceiling architecture for Engine C is a hybrid neuro-symbolic regulatory-reasoning engine:** a knowledge-graph of garments → materials → chemicals → regulations → numeric thresholds (encoded in RDF/JSON-LD with the UNTP DPP vocabulary and CIRPASS-2 textile ontology); a deterministic rules layer for the ~85+ hard-coded numeric thresholds (CPSIA 100 ppm lead, AB 1817 100→50 ppm TOF, REACH Entry 72's 33 CMR limits, OEKO-TEX Class I); a calibrated multi-label CatBoost-v1.2 + classifier-chain ML layer trained on CPSC SaferProducts recalls + EU Safety Gate + RAPEX + a synthetic Carter's-style garment population for the *uncertain* cases; a ChemBERTa-2/MolFormer chemical-language-model featurization head; an LLM-RAG layer over a vectorized regulatory corpus for explainability; and emission of a **W3C Verifiable-Credential-signed, GS1-Digital-Link-v1.4.0-encoded Digital Product Passport** that resolves through a CIRPASS-2-compatible resolver. SHAP per-prediction explanations feed the passport's "why-flagged" panel.
- **The single combination that maximizes wow-factor and commercial deployability:** encode every numeric threshold from Part 1 as deterministic Drools/Python-rules with cited statutory references; train the ML *only on the residual ambiguous cases* (unknown supplier, missing test data, partial composition); express the output as a UNTP-conformant DPP JSON-LD credential signed with a DID (did:web); embed the GTIN-based GS1 Digital Link in the QR (<https://id.gs1.org/01/{GTIN}/21/{serial}>)); and demonstrate eco-modulation savings against the Refashion 2024–2026 grid (€0.07/€0.70 durability bonus, €0.30/€0.03 certification bonus, €500–€1,000/tonne recycled-material bonus per the Refashion 2024 Guide to Eco-Modulation) plus California SB 707's per-unit eco-modulated fee (Landbell USA selected as PRO by CalRecycle on 27 February 2026; producer registration begins 1 July 2026).
- **Commercial ceiling:** third-party CPSIA panels run ~\$380–\$3,000/SKU; full AAFA RSL screening ~\$800–\$2,500/SKU; ZDHC MRSL Level 1 ~\$1,500–\$4,000/formulation; per Allianz Global Corporate & Specialty's December 2017 report "Product Recall: Managing The Impact of the New Risk Landscape," "the average cost of a significant incident is in excess of US\$12 million (€10.5 million), with the costs from the largest events far exceeding this total" — where "significant" is defined as product-recall claims ≥ €5m; CPSIA civil penalties cap at **\$120,000/violation** and **\$17,150,000 aggregate** (CPSC Federal Register notice, 86 FR 68244, 1 Dec 2021); per MarketsandMarkets's press release of 4 February 2025, "The Digital Product Passport Market is slated to expand from USD 185.9 million in 2024 to USD 1,780.5 million by the year 2030 at an impressive CAGR of 45.7%"; The Business Research Company alternatively projects \$1.35B (2025) → \$5.64B (2030) at 33.3%

CAGR; the textile-TIC market sits at ~\$4.5B (2025) → \$8.1B (2034) at 6.79% CAGR (360 Research Reports). The "Recyclability-Passport-as-a-Service" pitch is a credible standalone Series A play against EON (\$20.5M total, \$10M Series A led by Imaginary Ventures, 17 Feb 2022), TrusTrace (\$30M total, \$24M Series B led by Circularity Capital, 18 Jan 2024), Retraced (€15M Series A led by Partech, 10 Sept 2024), Sourcemap (\$20M Series B led by Energize Ventures, 6 June 2023), Worldly (\$54M total, \$50M Series B co-led by Silversmith Capital Partners and Galvanize Climate Solutions, announced 27 April 2022 per SGB Media), Circularise (€11M Series A from Brightlands/Asahi Kasei/Neste/4impact, Nov 2022; Teijin Group strategic round 5 Dec 2024), and Assent (\$497M total, \$350M Series D led by Vista Equity Partners, 6 Jan 2022, \$1B+ valuation). (AGCS Global + 11)

Key Findings

1. **The U.S./EU regulatory universe for children's stretch apparel is dense, numeric, and machine-encodable.** Every applicable rule reduces to a chemical-vs-threshold-vs-matrix triple. Engine C's defensible advantage is encoding all of them as hard, traceable, citation-linked rules — not as ML guesses.
 2. **All training-data sources required for the ML layer are free and public** (CPSC SaferProducts REST API, EU Safety Gate XML/Excel, EPA CompTox Dashboard, ECHA Annex XVII, PubChem). No paid feeds are required for a defensible v1.
 3. **The DPP standard stack has converged in 2025–2026:** ESPR 2024/1781 framework + CIRPASS-2 textile ontology + UNTP-style W3C Verifiable Credentials + GS1 Digital Link v1.4.0 over QR (ISO/IEC 18004) + did:web identifiers. Building to this stack today is not bleeding-edge — it is the emerging mainstream that 2027–2028 mandates will require.
 4. **Eco-modulation is the lever that monetizes the passport.** France Refashion's 2024 grid pays bonuses up to €1.00 per T-shirt against a ~€0.04 baseline eco-fee; California SB 707 (Governor Newsom signed 28 Sept 2024) made California the first U.S. state with mandatory textile EPR, with PRO Landbell USA selected on 27 Feb 2026 and per-unit eco-modulated fees to follow.
 5. **Demonstrating an automated DPP that survives ESPR's still-being-drafted textile delegated act (adoption expected 2027, application no earlier than 2028) is a credible "compliance-as-a-service" venture thesis** — the DPP software market is forecast to grow at a 33–46% CAGR from a 2024 base of **USD 185.9M (MarketsandMarkets, 4 Feb 2025)** to **USD 1,780.5M by 2030** at a 45.7% CAGR, or from \$1.35B (2025) to \$5.64B (2030) at 33.3% under The Business Research Company's broader scope. (PR Newswire)
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Details

PART 1 — THE FULL REGULATORY UNIVERSE (with exact thresholds, statutory citations, and effective dates)

A. United States — Federal (CPSC, 15 U.S.C. § 2051 et seq. / 16 CFR)

Consumer Product Safety Improvement Act (CPSIA) of 2008, Pub. L. 110-314 (15 U.S.C. § 1278a; § 2057c). Applies to any product "designed or intended primarily for children 12 years of age or younger."

- **Total lead content in substrate: ≤ 100 ppm (0.01%),** CPSIA § 101(a); codified at 16 CFR § 1500.87 and § 1252.1(a). 16 CFR § 1500.91(d)(7) determines that natural and manufactured-fiber textiles (dyed or undyed, but not screen-printed or after-treated) do **not** require third-party testing for lead — Engine C should encode this as a default exemption. (eCFR) (CPSC)
- **Lead in paint/surface coatings: ≤ 90 ppm (0.009%),** 16 CFR § 1303. (CPSC)
- **Eight prohibited phthalates at $> 0.1\%$ (1,000 ppm) in any accessible component part of children's toys or child-care articles** (16 CFR § 1307; CPSIA §§ 108(a) and 108(b)(3)): DEHP, DBP, BBP, DINP, DIBP, DPENP, DHEXP, DCHP. Children's sleepwear and bibs are *child-care articles* and are therefore in scope. (eCFR) (CPSC)
- **Third-party testing by CPSC-accepted laboratory + issuance of a Children's Product Certificate (CPC) + tracking labels:** § 14(a)(2) of CPSA; 16 CFR §§ 1107, 1109, 1110.
- **Civil penalties** (15 U.S.C. § 2069): **\$120,000 per violation; \$17,150,000 for any related series**, effective for violations after 1 Jan 2022 (86 FR 68244, 1 Dec 2021). Next CPSC five-year adjustment due 1 Dec 2026 effective 1 Jan 2027. The 2026 OMB-administered general civil-penalty inflation update was cancelled (M-26-11, 17 Apr 2026) due to the missing October 2025 CPI-U after the late-2025 government shutdown. (Foley & Lardner LLP + 2)

Flammability — 16 CFR Part 1610 (Standard for the Flammability of Clothing Textiles).

- Specimen 50 mm $\times$ 150 mm; 45° angle; 16 mm (5/8 in) standardized flame for 1 s; time to traverse 127 mm (5 in) measured. (QIMA)
- **Class 1 (Normal):** plain ≥ 3.5 s; raised ≥ 7 s. (CPSC)
- **Class 2 (Intermediate):** raised-surface only, 4.0–7.0 s with surface flash base burn (SFBB). (CPSC)
- **Class 3 (Rapid & Intense):** plain < 3.5 s, OR raised < 4.0 s with base burn — **prohibited** from clothing. (CPSC)
- Tested before and after refurbishing per § 1610.6.

Flammability — 16 CFR Part 1615 (sizes 0-6X) and Part 1616 (sizes 7-14) — children's sleepwear.

- Specimens 89 mm × 254 mm (3.5 × 10 in); vertically suspended; flame impinged 3 s; **char length measured per § 1615.4 / § 1616.5.**
- **Pass criterion: average char length of five specimens ≤ 7 in (178 mm); no single specimen char length > 10 in (254 mm).** (Qima)
- Test as produced (or after one laundering) **and** after 50 launderings per AATCC 124-1996.
- Exemption: tight-fitting sleepwear (and the 0-9 month subset) meeting dimensional limits in § 1615.1(c) / § 1616.2(c). **Stretch garments containing elastane often qualify — exactly REWEAR-FUSED's competitive surface.** (Qima)

B. United States — State PFAS-in-Textiles Laws

State	Statute	Effective	Scope	Threshold / Mechanism
California	AB 1817 ("Safer Clothes and Textiles Act"); H&S Code § 108970-108971 (signed 29 Sept 2022) + AB 347 (2024)	1 Jan 2025	New textile articles incl. apparel, accessories, handbags, draperies	Intentionally added PFAS prohibited; TOF ≤ 100 ppm from 1 Jan 2025, ≤ 50 ppm from 1 Jan 2027. (CA) Outdoor severe-wet exempt with "Made with PFAS Chemicals" disclosure until 1 Jan 2028. (Buchalter) Certificate of Compliance required. DTSC registration & enforcement from 1 July 2030; penalties ≥ \$10,000/violation (Morgan Lewis)
New York	S6291/A995 → Env. Cons. Law § 37-0121 (Bill S1322/A994)	1 Jan 2025	Apparel; outdoor severe-wet exempt until 1 Jan 2028	Intentionally added PFAS prohibited; DEC must set numerical threshold by 1 Jan 2027 (Morgan Lewis)
Maine	LD 1503 (2021) → LD 1537 (16 Apr 2024; P.L. 2023, c. 630) , 38 MRSA § 1614	Phased: 1 Jan 2026 (textile articles, juvenile products, cookware, cosmetics, ski wax, upholstered furniture); 1 Jan 2029 (artificial turf, outdoor severe-wet apparel); 1 Jan 2032 (all other); 1 Jan 2040 (HVAC/refrigerants)	Intentionally added PFAS	Total ban subject to "Currently Unavoidable Use" exemption (5-year, renewable)

State	Statute	Effective	Scope	Threshold / Mechanism
Minnesota	Amara's Law , Minn. Stat. § 116.943 (2023)	1 Jan 2025 (Phase I: 11 categories incl. juvenile products, fabric treatments, textile furnishings); reporting due 15 Sept 2026 via PRISM 1.0; 1 Jan 2032 (Phase III: all products)	Intentionally added PFAS	Total ban with CUU exemption; MPCA final reporting rule adopted 8 Dec 2025
Colorado	HB22-1345; HB23-1215	1 Jan 2025 (outdoor severe-wet with disclosure); 1 Jan 2028 (all severe-wet outdoor + textile articles)	Intentionally added PFAS	Total ban
Washington	Safer Products for WA (RCW 70A.350) Cycle 1.5 (apparel/gear)	Phased reporting 2025– 2026; restrictions follow rulemaking	CHCC; PFAS in apparel	Notification + restrictions per rulemaking
Rhode Island	H 7356 / S 2152 (2024)	1 Jan 2027 (textile articles ban)	Intentionally added PFAS	Ban
Vermont	S.25 (2024)	1 July 2026 (apparel ban)	Intentionally added PFAS	Ban
Connecticut	Public Act 24-59 + DEEP Final Labeling Order Dec 2025	Labeling from 1 July 2026; restrictions 2028– 2032	Intentionally added PFAS	Phased ban + labeling ("Contains PFAS"/"Made with PFAS Chemicals"/etc.)
Oregon	Toxic-Free Kids Act (HB 3043, 2015)	Ongoing	Children's products	CHCC reporting/disclosure

C. California Proposition 65 (H&S § 25249.5; OEHHA Safe Harbor Levels per 27 CCR §§ 25705/25709/25805)

- **Lead: NSRL 15 µg/day (carcinogen); MADL 0.5 µg/day (reproductive)** — MADL is the binding figure for textiles.
- **Cadmium (oral): MADL 4.1 µg/day.**

- DEHP: NSRL 310 µg/day; MADL 410 µg/day.
- DBP: MADL 8.7 µg/day; BBP: MADL 1,200 µg/day.
- DINP: NSRL 146 µg/day.
- Formaldehyde (inhalation): NSRL 40 µg/day.
- BPA (dermal): MADL 3 µg/day.
- Full table maintained by OEHHA at oehha.ca.gov/proposition-65/general-info/proposition-65-no-significant-risk-levels-nsrls-and-maximum-allowable-dose-levels-madls. Compliance for businesses with ≥ 10 employees selling in CA.

D. European Union — REACH (Regulation (EC) 1907/2006) Annex XVII

- **Entry 43 (azo dyes):** azo colorants that, by reductive cleavage, may release any of the **22 carcinogenic aromatic amines in Appendix 8** at > **30 mg/kg (0.003%)** in textile/leather articles in direct/prolonged skin contact are prohibited. Test methods: EN ISO 14362-1, EN ISO 14362-3 (p-aminoazobenzene), EN ISO 17234-1/2 (leather). The 22 amines include benzidine, o-toluidine, 4-aminobiphenyl, 4-chloro-o-toluidine, 2-naphthylamine, 4,4'-oxydianiline, 4,4'-methylene-bis-(2-chloroaniline), 3,3'-dichlorobenzidine, 3,3'-dimethoxybenzidine, 3,3'-dimethylbenzidine, p-cresidine, 4,4'-thiodianiline, o-anisidine, 4-aminoazobenzene, 2,4-toluenediamine, 4,4'-methylene-bis-aniline, 4-methyl-m-phenylenediamine, 2,4,5-trimethylaniline, o-aminoazotoluene, 2-amino-4-nitrotoluene, p-chloroaniline, 2,4-diaminoanisole. Appendix 9 lists banned dyes at > 0.1%.
- **Entry 47 (Chromium VI in leather):** ≤ 3 mg/kg in leather articles or parts in skin contact.
- **Entry 50 (PAHs):** 8 PAHs (BaP, BeP, benzo[a]anthracene, chrysene, benzo[b]/[j]/[k]fluoranthene, dibenzo[a,h]anthracene) ≤ 1 mg/kg for skin-contact articles; ≤ 0.5 mg/kg for toys and childcare articles.
- **Entry 72 (33 CMR Cat. 1A/1B substances in clothing, related accessories, similar-skin-contact textiles, footwear) — Commission Regulation (EU) 2018/1513; effective 1 Nov 2020.** Key individual limits: (SATRA)
 - **Formaldehyde:** ≤ 75 mg/kg (general); the historical 300 mg/kg derogation for jackets/coats/upholstery expired 1 Nov 2023. (Measurlabs) (SGS)
 - **Quinoline:** ≤ 50 mg/kg.
 - **Three disperse dyes (Disperse Blue 1, Disperse Orange 37/76/59, Disperse Yellow 3):** ≤ 50 mg/kg each (SATRA TM459:2024). (SATRA)
 - **Benzene:** ≤ 5 mg/kg.
 - **Cadmium, Lead, Chromium VI, Arsenic:** ≤ 1 mg/kg each.
 - **PAHs (8):** ≤ 1 mg/kg each.
 - **Three chlorotoluenes (α,α,α,4-tetrachlorotoluene; p-chlorobenzotrichloride; etc.):** ≤ 1 mg/kg each.
 - **Phthalates Σ(DEHP + DBP + BBP + DIBP):** ≤ 1,000 mg/kg (0.1%).

- **N,N-dimethylacetamide; N,N-dimethylformamide; N-methyl-2-pyrrolidone:** ≤ 3,000 mg/kg each.
- **1,2-benzenedicarboxylic acid, di-C6-8-branched alkyl esters, C7-rich:** ≤ 1,000 mg/kg.
- **Entry 77 (Formaldehyde in articles) — Commission Regulation (EU) 2023/1464 (14 July 2023, OJ L 180/12); applies 6 Aug 2026** (6 Aug 2027 for vehicle interiors).
 - **0.062 mg/m³** for wood-based articles and furniture (test-chamber emissions per Appendix 14). (EaseCert)
 - **0.080 mg/m³** for other articles (textiles, leather, plastics).
 - Articles within scope of Entry 72 are explicitly **exempt** from Entry 77 — Entry 72's 75 mg/kg content limit binds finished textiles. (UL Solutions)

E. EU Textile Strategy, ESPR, and the Digital Product Passport

- **Regulation (EU) 2024/1781 (ESPR)**, in force 18 July 2024, full application 19 July 2026; replaces Directive 2009/125/EC.
- **ESPR Working Plan 2025-2030** (adopted 16 April 2025) names **textiles and apparel as a priority** — delegated act adoption indicatively **2027**; mandatory DPP earliest **2028** (after the minimum 18-month application window). (Circularise) (COSHI)
- **JRC Preparatory Study on Textiles** reached its third milestone in December 2025. (Regen Studio)
- **EU Central DPP Registry** goes live alongside ESPR full application on **19 July 2026** (per CIRPASS-2 / Fiegenbaum Solutions consolidated register, 22 April 2026). (Fiegenbaum Solutions)
- Projected textile-DPP data model (extrapolating Battery Reg (EU) 2023/1542 and CIRPASS-2 working docs): unique product identifier (GTIN + serial), material composition (fiber % + supplier disclosure), substances of concern (CAS-level for any SVHC > 0.1%), recyclability indicators (fiber-to-fiber recyclability, separability), care instructions, durability metrics (Martindale, color fastness, dimensional stability, tensile), supply-chain/traceability events (EPCIS-style), carbon footprint (cradle-to-gate per EN 15804/ISO 14040-44).

F. OEKO-TEX Standard 100 Edition 2025 (valid since 1 April 2025) — Class I (babies/young children ≤ 36 months)

The strictest of four product classes (I — babies; II — direct skin contact; III — no skin contact; IV — decoration). Class I key limits:

- **Formaldehyde: not detectable (< 16 mg/kg)** per ISO 14184-1 (vs. 75 mg/kg for Class II; 300 mg/kg for III/IV). (MOSSRIVER)
- **Extractable heavy metals (mg/kg, artificial saliva/acid sweat):** As 0.2; Pb 0.2; Cd 0.1; Cr 1.0; Cr(VI) 0.5 mg/kg detection limit; Co 1.0; Cu 25.0; Ni 1.0; Hg 0.02; Sb 30.0.

- **pH: 4.0–7.5.** (MOSSRIVER)
- **Pesticides (sum):** ≤ 0.5 mg/kg.
- **Pentachlorophenol (PCP):** ≤ 0.05 mg/kg; **TeCP:** ≤ 0.05 mg/kg.
- **Banned azo arylamines:** ≤ 20 mg/kg each (stricter than REACH 30 mg/kg).
- **Phthalates:** $\Sigma \leq 0.05\%$ for Class I.
- **APEOs (NPEO + OPEO):** ≤ 250 mg/kg.
- **Total PFAS analysis added in Edition 2024;** PFOA ≤ 0.025 mg/kg; PFOS ≤ 1.0 $\mu\text{g}/\text{m}^2$.
(MOSSRIVER)
- **Color fastness to saliva and perspiration:** no migration permitted.

G. EU General Product Safety Regulation (GPSR), Regulation (EU) 2023/988

Applies from **13 December 2024**; replaces Directive 2001/95/EC. Imposes risk-assessment, technical-documentation, EU Responsible Person (Art. 16), traceability labeling, Safety Gate notification, and online-marketplace cooperation obligations on all non-food consumer products including textiles. (Centexbel) (Eurofins Scientific)

H. Other Encodable Standards

- **EN 1811 (nickel release):** ≤ 0.5 $\mu\text{g}/\text{cm}^2/\text{week}$ for prolonged skin contact; ≤ 0.2 $\mu\text{g}/\text{cm}^2/\text{week}$ for piercing posts; codified into REACH Annex XVII Entry 27.
- **AAFA Restricted Substances List 25th Edition (2024):** 250+ substances; voluntary industry standard.
- **ZDHC Manufacturing Restricted Substances List v3.1 (2022):** input-chemistry standard.
- **bluesign® System Substances List (BSSL):** input-chemistry positive list.

PART 2 — DATA: PUBLIC DATASETS AND APIs FOR THE COMPLIANCE CLASSIFIER

A. CPSC SaferProducts Recall REST API

- **Endpoint:** (<https://www.saferproducts.gov/RestWebServices/Recall?format=json>) (XML alternative). (CPSC)
- **Format:** JSON/XML; ~9,000+ recall records (~50 MB full pull). (Apify)
- **Update frequency:** real-time as recalls are issued.
- **Fields:** (RecallID), (RecallNumber), (RecallDate), (Description), (URL), (Title), (ConsumerContact), (LastPublishDate), (Hazards[] .HazardTypeID/Name), (Images[]), (Injuries[]), (Inconjunctions[]), (Manufacturers[]), (ManufacturerCountries[]), (ProductUPCs[]), (Products[] .CategoryID/Name/Description/Model/Type/NumberOfUnits), (Remedies[]), (Retailers[]). (Compjour)

- **License:** U.S. Government work, public domain.
- **Programmer Guide:** (cpsc.gov/s3fs-public/RecallRetrievalWebServicesProgrammersGuide20180917.pdf) (v1.4).
- **Use in ML:** Positive labels for "non-compliant children's apparel" — filter Products.CategoryID for apparel/children categories; tokenize Hazards.Name and Description to extract chemical entities ("lead paint", "phthalates"), choking, flammability classifications.

B. EU Safety Gate / RAPEX

- **Portal:** (<https://ec.europa.eu/safety-gate-alerts/>).
- **Weekly XML/Excel reports:** (https://ec.europa.eu/safety-gate-alerts/api/download/weeklyReport/detail/xml/{week_id}?language=en).
- **Volume:** ~2,000 notifications annually; ~80,000 historical records. ([Bund](#))
- **Fields:** Alert number, Notifying country, Product category (toys, motor vehicles, **clothing-textiles-fashion**), Risk type (chemical, choking, injury, microbiological), Risk description, **Chemical detected** (phthalates, formaldehyde, AZO dyes, lead, cadmium), Brand, Model, Country of origin, Measures taken, Image URLs.
- **License:** Re-use permitted with attribution under Commission Decision 2011/833/EU.

C. EU RASFF (food/feed) — adjacent design pattern (webgate.ec.europa.eu/rasff-window/)).

D. PFAS-in-Textiles Testing Datasets

- **Toxic-Free Future "Mind the Store"** retailer testing (toxicfreefuture.org): TOF and targeted-PFAS results for hundreds of products including children's brands.
- **Notre Dame / Graham Peaslee group:** total-fluorine PIGE studies, *Environmental Science & Technology Letters* multiple papers 2018–2024, supplementary product-level CSV.
- **Silent Spring Institute "Detox Me"** datasets: (silentspring.org/research-area/consumer-products).

E. OEKO-TEX Buying Guide

- **URL:** (oeko-tex.com/en/buying-guide). Public registry of certificates; queryable by company, country, Standard 100 class. ([OEKO-TEX](#))

F. EPA CompTox Chemicals Dashboard

- **URL:** (comptox.epa.gov/dashboard) (v3.0.5+).
- **API:** (<https://api-ccte.epa.gov/>) (CCTE Resources API), free key.

- **PFAS lists:** PFASMASTER (14,735 PFAS as of Aug 2022; expanding), PFASSTRUCT, PFASOECD, PFASKEMI, PFASTRIER, EPAPFASRL. ([PubMed Central](#))
- **Fields:** DTXSID, CASRN, SMILES, InChI, predicted phys-chem, ToxCast bioassay, OPERA/TEST predictions.
- **License:** U.S. Government work.

G. ECHA — Annex XVII restrictions (echa.europa.eu/substances-restricted-under-reach); **Candidate List of SVHCs**; **CoRAP**; **C&L Inventory**.

H. PubChem

- **API:** (pubchem.ncbi.nlm.nih.gov/rest/pug/) — chemical-structure retrieval for ChemBERTa/MolFormer featurization.

I. Higg / Cascal Material Sustainability Index (MSI)

- **Worldly platform** (formerly Higg Co): paid; Higg MSI 3.7 (2024 update). Reference only.

J. Synthetic Garment Composition (for the classifier's negative population)

- Generate ~10,000 SKUs via stratified sampling: size (0-14), fiber composition (cotton/polyester/elastane), color (azo-dye risk proxy), country of origin (CPSC recall-rate prior), supplier tier (OEKO-TEX status), washcare label. Inject ~3% non-compliance to match empirical CPSC apparel-recall base rate.

PART 3 — THE MAXIMUM-CEILING ML ARCHITECTURE

A hybrid neuro-symbolic stack.

1. **Knowledge Graph (Neo4j or RDF/SPARQL via Apache Jena):** Garment → MaterialBOM → Substance(CAS) → RegulationEntry → Threshold. 85+ explicit thresholds from Part 1 are RDF triples queryable via SPARQL.
2. **Deterministic Rules Layer (Python [python-rule-engine](#) or Drools via py4j):** With complete composition data, rules alone determine compliance — e.g., [IF age_group ≤ 12 AND lead_substrate_ppm > 100 THEN flag\(reg="CPSIA §101", severity="HARD_FAIL"\)](#). ML is *not consulted* in deterministic cases. This is essential for judge/regulator trust.
3. **ML Layer (residual uncertain cases):**
 - **Model:** CatBoost v1.2 (native categorical support; outperforms LightGBM/XGBoost on tabular mixed types with supplier names, country, color as categoricals) as the primary multi-label classifier, wrapped in **scikit-learn** ([ClassifierChain](#)) to respect

label dependencies (e.g., AB 1817 positive correlates with Cal-EPR risk; OEKO-TEX absence elevates REACH risk).

- **Imbalance handling:** focal loss (CatBoost `loss_function="Logloss"` + class weights) or `imbalanced-learn` SMOTE-NC. Positives <5%.
- **Calibration:** isotonic regression on held-out fold; report **PR-AUC not ROC-AUC** because positives are rare. Targets: per-label PR-AUC ≥ 0.85 , macro PR-AUC ≥ 0.80 , Brier ≤ 0.05 .
- **Explainability:** SHAP TreeExplainer per prediction; top-5 SHAP features rendered into the DPP's "why-flagged" panel.

4. Chemical-Structure Featurization:

- **RDKit Morgan fingerprints** (radius=2, 2048 bits).
- **ChemBERTa-2** (`DeepChem/ChemBERTa-77M-MTR`) embeddings (768-dim).
- **MolFormer** (IBM `ibm/MolFormer-XL-both-10pct`) embeddings, pretrained on 1.1B molecules — for generalization to novel SoCs.

5. **LLM-RAG Regulatory Reasoner:** LlamaIndex/LangChain RAG over a vectorized corpus of (i) statutory text (CPSIA, REACH, AB 1817, Refashion grid), (ii) ECHA guidance, (iii) CPSC FAQs, (iv) OEKO-TEX criteria catalog. Backend: `gpt-4o-mini` or `claude-3.5-haiku` for cost; embeddings via `text-embedding-3-large`. **The LLM never decides compliance** — it explains decisions and parses unstructured supplier datasheets/recall descriptions.

6. **NER for Chemicals:** scispacy `en_ner_bc5cdr_md` or ChemListem.

Why this beats a single LightGBM model: a single tabular model cannot represent the *hard* deterministic structure of statutory thresholds, and judges/regulators will (correctly) distrust a probabilistic estimate of "is lead < 100 ppm". Engine C's hybrid stack uses ML *only where it belongs*: filling missing fields, predicting risk from supplier history, classifying free-text descriptions, and matching unknown substances by structure.

PART 4 — THE DIGITAL PRODUCT PASSPORT: STANDARDS AND IMPLEMENTATION

A. Required DPP data fields (textile, projected from CIRPASS-2 + Battery Reg + JRC Dec 2025 study)

- Unique product identifier (GTIN-14 + **serial 21** per GS1 Digital Link).
- Manufacturer identity (DID + registered legal entity).
- **Material composition** (fiber percentages with CAS for polymer additives).
- **Substances of concern** (any SVHC > 0.1% w/w; chemical name + CAS + concentration).
- Recyclability indicators (mono-material fraction, separability of trims, enzymatic-degradability flag — REWEAR-FUSED's unique surface).
- Care instructions (ISO 3758 symbols).

- Durability metrics (Martindale ISO 12947, color fastness ISO 105, dimensional stability ISO 5077, tensile ISO 13934-1).
- Supply-chain / traceability events (GS1 EPCIS 2.0 commissioning, shipping, receiving).
- Carbon footprint (cradle-to-gate; EN 15804 / ISO 14040-44).
- Eco-modulation evidence (claims that justify Refashion bonus or SB 707 fee reduction).

B. GS1 Digital Link v1.4.0 — Encoding the QR

- **URI syntax:** `(https://id.gs1.org/01/{14-digit-GTIN}/21/{serial})` (optional `/10/{batch}`), `(?17={expiry})`. (DEV Community)
- **Standards:** ISO/IEC 18004 QR Code; ISO/IEC FDIS 18975 (encoding-and-resolving identifiers over HTTP). v1.4 dropped "convenience alphas" — numeric AIs only. (Tracehub + 2)
- **Recommended QR:** Model 2, ECC Level M (15%); min 15 mm × 15 mm @ 150 dpi; quiet zone ≥ 4 modules. (DEV Community + 2)
- **Resolver:** brand-operated or `(id.gs1.org)` (GS1 Global Office resolver, open source). (GS1)
- **GS1 Sunrise 2027** requires POS systems in 48+ countries to accept 2D barcodes by year-end 2027 — bakes Digital Link QR adoption into retail. (DEV Community)

C. UNTP — UN Transparency Protocol (UN/CEFACT, UNECE Recommendation No. 49)

- **Specification:** `(untp.unece.org/docs/specification/)`.
- DPP issued as a **W3C Verifiable Credential 2.0** (JSON-LD), `(ProductPassport)` subject under UNTP @context. (Unece)
- Three building blocks: **Digital Product Passport** (claims), **Digital Conformity Credential** (third-party verification), **Digital Traceability Events** (EPCIS-style). (UNCTAD)
- Resolver-based "pull" model centered on product identifier. (Unece)
- Cryptographic signing via DID (**did:web** easiest start; did:ion for full decentralization).
- The credential's `(renderMethod)` property allows the same signed credential to render both as machine-readable JSON and as a human-readable HTML "passport page." (Unece)

D. CIRPASS-2 (May 2024 - Apr 2027; Horizon Europe; 49 partners)

- Textile DPP upper-level ontology (T3.2, led by GS1 in Europe). (Cirpass2)
- LCA ontology (maki). (Cirpass2)
- 13 pilots across textiles/electronics/tyres/construction. (Fraunhofer)
- **EU DPP Core Ontology requirements** published March 2025 — the de facto interoperability reference for ESPR delegated acts. (Fiegenbaum Solutions)
- **ISO/IEC JTC 5** on Digital Product Passports announced 20 April 2026 (DIN/DKE); substantive deliverables 2028+. (Fiegenbaum Solutions)

- **For:** tamper-evidence; multi-party trust without single host.
- **Against:** cost per write; complexity; permissioned consortium chains (Hyperledger Fabric) lack public verifiability.
- **Recommendation:** anchor only a **Merkle root of credential hashes** on a low-cost L2 (Polygon PoS) or the **European Blockchain Services Infrastructure (EBSI)** on a daily batch — preserves verifiability without per-credential cost. Credentials themselves live in IPFS or a CIRPASS-2-compliant registry. **EBSI is the politically defensible choice for an EU audience.**

F. Existing DPP textile platforms (competitive landscape)

- **EON** (formerly EON Group): Digital ID for fashion; partners with Coach, Mulberry, H&M, Pangaia, Chloé. \$20.5M total raised; \$10M Series A led by Imaginary Ventures (Natalie Massenet), 17 Feb 2022.
- **Avery Dennison atma.io** (NYSE: AVY): connected-product cloud launched 2021; powers many luxury/footwear DPP pilots. No separate startup funding.
- **TextileGenesis:** acquired by Lectra (Euronext: LSS) in 2023; terms undisclosed; Fibercoin token traceability.
- **TrusTrace** (Stockholm): \$30M total; \$24M Series B on 18 Jan 2024 led by Circularity Capital (with Industrifonden, Fairpoint). [Dealroom.co](#)
- **Retraced** (Düsseldorf): €15M Series A 10 Sept 2024 led by Partech.
- **Circularise** (The Hague): €11M Series A Nov 2022 (Brightlands VP, Asahi Kasei, Neste, 4impact); Teijin Group strategic investment 5 Dec 2024 (amount undisclosed).
- **Sourcemap:** \$20M Series B 6 June 2023 led by Energize Ventures (with E14 Fund).
- **Worldly** (formerly Higg Co): per SGB Media's 27 April 2022 report, "Higg Co., the technology company behind the Higg Index platform, announced its \$50 million Series B funding round" co-led by Silversmith Capital Partners and Galvanize Climate Solutions (with Titan Grove, Buckhill Capital); \$54M total raised. [SGB Media](#)

PART 5 — COMMERCIAL & REGULATORY-STRATEGY CEILING

A. Per-SKU children's apparel testing costs (industry-quoted; major TIC firms do not publish list prices)

- **CPSIA panel** (lead substrate + lead in paint + 8 phthalates): **\$380-\$3,000/SKU** depending on materials. AQI Service (Mar 4, 2026) quotes **\$380-\$500** for a standard CPSIA-only panel with a 7-10 business-day turnaround; JJR Lab cites **\$1,500-\$3,000** for full CPC certification. [AQI Service](#)

- **PFAS targeted LC-MS/MS (EPA Method 1633, 40 compounds):** ~\$250–\$600/sample commercially; combustion-IC total fluorine ~\$150–\$350/sample. Bureau Veritas (bvna.com/environmental-ih-laboratories/services/pfas-testing) is accredited for EPA Method 1633. Pintas & Mullins cites \$300–\$600/sample at certified labs (for water testing — textile costs comparable). (BVNA) (Pintas & Mullins)
- **Full AAFA RSL screening (250+ analytes):** \$800–\$2,500/SKU (industry-quoted).
- **ZDHC MRSL v3.1 Level 1 conformance:** \$1,500–\$4,000/formulation.
- **16 CFR 1610 (45° flammability):** \$75–\$200/sample.
- **16 CFR 1615/1616 (children's sleepwear; 50-wash conditioning):** \$800–\$1,800/SKU.

A children's brand launching 500 new SKUs/season faces **\$0.5M–\$2M/year in chemical compliance testing alone**. Engine C's positioning: the DPP enables risk-based testing reduction — if the passport demonstrates compositional certainty + supplier OEKO-TEX history + clean recall record, the brand can target testing on residual risk only.

B. Cost of a recall

- Per Allianz Global Corporate & Specialty's December 2017 report "Product Recall: Managing The Impact of the New Risk Landscape," "the average cost of a significant incident is in excess of US\$12 million (€10.5 million), with the costs from the largest events far exceeding this total. Over 50% of losses arise from 10 incidents" — analyzing 367 claims across 28 countries / 12 sectors 2012–H1 2017, where "significant" means claims ≥ €5m (~\$5.8m). (AGCS Global + 3)
- FMI / GMA / GS1 US / Deloitte (per Nationwide, 2025): the average food recall costs **\$10M** (apparel-specific figure not separately disclosed by Allianz or Sedgwick). (Nationwide)
- Sedgwick Brand Protection 2025 U.S. State of the Nation Recall Index: 3,232 recalls across five major industries in 2024; Sedgwick has "managed 8,000+ of the most sensitive and time-critical recalls in 150+ countries." (Nationwide) (Sedgwick)
- **CPSIA civil penalties (15 U.S.C. § 2069):** per CPSC Federal Register notice 86 FR 68244 (1 Dec 2021), "the adjusted maximum amounts are \$120,000 for each violation, and \$17,150,000 for any related series of violations. These new amounts apply to violations that occur after January 1, 2022." Pending CAP Act (introduced 25 Jan 2024) would raise per-violation cap to \$250,000 and eliminate the aggregate cap; not enacted as of mid-2026. (Federal Register + 2)

C. Eco-modulation: how the passport directly cuts producer fees

California SB 707 (Responsible Textile Recovery Act of 2024), signed by Governor Newsom 28 Sept 2024; Pub. Res. Code §§ 42985–42985.27.

- **PRO: Landbell USA** selected by CalRecycle on **27 February 2026** (per Anthesis 2026, rev-log.com 2026). (Rev-Log)
- **Producer registration:** from **1 July 2026** (covered producers = > \$1M global sales selling apparel/textiles in California). (Americancirculartextiles)
- **Needs assessment:** due March 2027. (Rev-Log)
- **Plan / regulations:** effective no earlier than **1 July 2028**; full enforcement upon plan approval or **1 July 2030** whichever earlier. (Rev-Log) (Holland & Knight)
- **Penalties:** per CalMatters, "administrative civil penalties for a violation of the programs requirements, not to exceed \$10,000 per day, or not to exceed \$50,000 per day for an intentional or knowing violation." (Digital Democracy)
- **Eco-modulation mechanism:** per Holland & Knight's October 2024 analysis, "the intent of the eco-modulated fee is to incentivize collection plans that facilitate the achievement of reusing, repairing and recycling covered products through reduced fees, while disincentivizing production of materials incongruent with that goal by charging producers higher fees." Specific fee schedule **not yet published**; Engine C's value proposition: **the passport is the documentation that justifies the lower fee.** (Holland & Knight)
- **PFAS handling explicitly required in the PRO plan.** (Greenstitch)

France — Refashion (AGEC Law Art. 62) eco-modulation grid 2024-2026 (Refashion 2024 Guide to Eco-Modulation, May 2024):

- **Baseline eco-contribution:** per H       Daret, Refashion brands division director, quoted in Maison du Savoir-Faire, "All new products put on the market must pay eco-contributions, the amount of which is set every year, averaging just under 4 centimes per item for 2024." (Maisondusavoirfaire)
- **Durability bonus:** "the bonus paid for each style corresponds to the multiplication of a reference amount (  0.07 or   0.70 depending on the volume placed onto the market) and a factor specific to each of the 10 [product] categories" (Refashion 2024 Guide).
- **Environmental certification bonus:** "The bonus paid for each style is   0.30 or   0.03 depending on the volume placed on the market and the product category (Clothing, Household linen, Footwear)." Eligible labels: Ecocert ERTS Level 2, OEKO-TEX MADE IN GREEN, Bluesign, Fairtrade Textile, EU Ecolabel, Demeter, GOTS, Bior  . (Refashion) (Refashion)
- **Recycled-material incorporation bonus:** "Raw materials from the recycling of post-consumer CHF textile waste collected or funded by an approved producer responsibility organisation for CHF textile management:   1,000/tonne"; "Recycled raw materials coming from open loop recycling of waste collected or funded by an approved producer responsibility organisation excluding food-grade plastic resin:   500/tonne." (Refashion)
- **Maximum cumulative example (Refashion):** "a T-shirt paying an eco-contribution of 2 centimes could receive up to a one-euro bonus." (Maisondusavoirfaire)

- **Malus from 1 January 2025:** per Refashion, "On 1st January 2025, a new eco-modulation penalty was introduced. It establishes two penalties for finished products that contain elements that affect their recyclability... Electronic or electrical components in clothing, household linen and footwear (excluding RFID-type traceability devices). The amount of the penalty corresponds to a percentage of the eco-fee amount for the product." (Refashion)
- **Proposed ultra-fast-fashion malus:** per WWD/Sourcing Journal (Nov 2025), the parallel anti-ultra-fast-fashion Senate bill "could eventually bring per-garment penalties (up to 10 euros by 2030) and an ad ban" — not yet law. (WWD)

D. Market size for compliance/traceability/DPP software

- **DPP software market:** estimates vary by scope:
 - **MarketsandMarkets** (PR Newswire, 4 Feb 2025): "The Digital Product Passport Market is slated to expand from USD 185.9 million in 2024 to USD 1,780.5 million by the year 2030 at an impressive CAGR of 45.7%." (PR Newswire)
 - **The Business Research Company / Research and Markets:** \$1.35B (2025) → \$1.8B (2026) → \$5.64B (2030) at 33.3% CAGR. (Research And Markets)
 - **Fortune Business Insights:** \$294M (2025) → \$4,545M (2034) at 35.56% CAGR. (Fortune Business Insights)
 - **Market Research Future:** \$1.862B (2024) → \$8.783B (2035) at 15.14% CAGR. (Market Research Future)
 - **360iResearch:** \$342M (2025) → \$663M (2032) at 9.92% CAGR. (360iResearch)
 - **Coherent Market Insights:** \$240.1M (2025) → \$1,273.2M (2032) at 26.9% CAGR. (Coherent Market Insights)
- **Global TIC market (overall):** per Global Market Insights, "the testing, inspection and certification (TIC) services market size was estimated at USD 256.9 billion in 2024 and is expected to grow at a CAGR of 5.2% between 2025 and 2034... Top 5 players in this market include SGS, Bureau Veritas, Intertek, TÜV SÜD, Eurofins, which collectively held a market share of 9.1% in 2024." Astute Analytica (11 Feb 2026) projects "US\$ 541.12 Billion by 2035." (Global Market Insights + 2)
- **Textile-TIC sub-market:** per 360 Research Reports, "\$4.5B (2025) → \$8.1B (2034) at 6.79% CAGR"; "SGS holds approximately 21% of total market share with over 2,600 textile testing facilities across 65 countries... Intertek Group (UK): Maintains around 18% of market share." Valuates Reports projects textile-TIC at US\$10.74B by 2030. Mordor Intelligence projects >7.5% CAGR 2024–2029. (360researchreports + 3)
- **2024 TIC firm revenues** (Aventis Advisors): SGS \$7.72B, Eurofins \$7.52B, Bureau Veritas \$6.98B, Intertek/DEKRA > \$4B each. **Intertek 2024 full-year results (4 Mar 2025):** "Our Softlines business delivered double-digit LFL revenue growth as we have seen an increase in ATIC investments by our clients in e-commerce, Risk-based Quality Assurance, end-to-end sustainability and in new products." **Bureau Veritas Q3 2024:** "The Softlines, Hardlines

& Toys segment (accounting for 46% of divisional revenue) delivered high-single-digit organic growth in Q3 2024." (Aventis Advisors + 2)

E. Standalone-venture positioning

REWEAR-FUSED's Engine C, repositioned as "**Recyclability-Passport-as-a-Service**" for children's and stretch apparel, sits at the intersection of three large adjacencies (textile TIC + traceability SaaS + EPR compliance). Defensible angle: **the only DPP platform that ships with a pre-encoded children's-apparel chemical-compliance brain**, not just a data container.

Comparable funding rounds confirming the thesis:

- **EON**: \$20.5M total; \$10M Series A by Imaginary Ventures, 17 Feb 2022. (Tracxn)
 - **TrusTrace**: \$30M total; \$24M Series B by Circularity Capital, 18 Jan 2024.
 - **Retraced**: €15M Series A by Partech, 10 Sept 2024. (Partech)
 - **Sourcemap**: \$20M Series B by Energize Ventures, 6 June 2023. (Sourcemap)
 - **Worldly**: \$54M total; \$50M Series B announced 27 April 2022 (Silversmith Capital Partners + Galvanize Climate Solutions co-leads). (SGB Media)
 - **Circularise**: €11M Series A Nov 2022 (Brightlands VP + Asahi Kasei + Neste + 4impact) + Teijin strategic Dec 2024. (Circularise) (Circularise)
 - **Assent** (adjacent compliance SaaS): \$497M total; \$350M Series D by Vista Equity Partners, 6 Jan 2022, \$1B+ valuation. (Tracxn) (FinSMEs)
-

Recommendations

Build now (hackathon deliverable, defensible for judges):

1. **Encode the deterministic rules layer first** (the threshold table in the appendix). Use Python (Rule) objects each carrying (statute_citation), (chemical_cas), (threshold_value), (unit), (matrix), (age_group), (effective_date), (severity). This is the unambiguous compliance core.
2. **Layer CatBoost v1.2 with classifier-chain** on top, trained on CPSC SaferProducts + Safety Gate + Mind-the-Store/Peaslee PFAS datasets + a 10,000-record synthetic Carter's-style population. Calibrate with isotonic regression. Report per-label PR-AUC and Brier.
3. **Emit a UNTP-conformant W3C Verifiable Credential** signed with a did:web key (the team's GitHub Pages domain). Include the GS1 Digital Link URI as the credential subject's (id). Render via the credential's (renderMethod) to an HTML "passport page" with the SHAP "why-flagged" panel.

4. **Generate the GS1 Digital Link v1.4 QR** using `(python-qrcode)` + a custom encoder targeting ISO/IEC 18975 path syntax `(/01/{GTIN}/21/{serial})`; resolver hosted on Vercel/Fly.io as a FastAPI service.
5. **Anchor the daily Merkle root of issued credentials on EBSI testnet** (or Polygon Amoy as a fallback). This is the tamper-evidence story for Cox investors without the cost of per-credential anchoring.

Scale (Series-A pitch arc): 6. **Target the AB 1817 + SB 707 + NY S6291 + ESPR-textile-DA stack** as the wedge — Carter's and the broader children's-apparel category needs all four solved by 2028. 7. **Quantify the Refashion bonus** for a representative SKU live during the demo: a Carter's recycled-elastane onesie hitting both durability + certification + recycled-material bonuses earns ~€1.00 vs. €0.04 baseline — a 25× return per garment. 8. **Price the SaaS as a percentage of testing-cost savings** (\$0.5M–\$2M/season per mid-tier brand) and EPR-fee savings — beats the alternative of being yet another "ESG platform."

Benchmarks/thresholds that should change the recommendation:

- If the ESPR textile delegated act publishes its data-field schema before pitch day, swap the projected schema for the official one immediately.
- If CalRecycle publishes Landbell USA's draft fee grid before pitch day, pre-compute the Carter's-SKU fee reduction live.
- If the CPSC's next penalty inflation adjustment (1 Jan 2027) lands, refresh the recall-cost slide.
- If OEKO-TEX Standard 100 Edition 2026 changes the total-fluorine PFAS limit, update the Class-I row.

Caveats

- The ESPR delegated act for textiles has **not been adopted** as of mid-2026; the data fields above are a projected schema based on CIRPASS-2 working documents, the Battery Regulation precedent, and the JRC Textile Preparatory Study third milestone (December 2025). Treat as "best available draft," not law.
- DPP software market size estimates from private analysts range from ~\$240M to \$1.78B (2025–26) and CAGRs from 9.9% to 45.7% — scope definitions vary. Cite ranges, not point estimates.
- Per-SKU TIC pricing is quote-based; the major firms do not publish list prices, so the ranges given are from secondary sources (AQI Service, JJR Lab, SUNY-ESF rate cards, Pintas & Mullins, Measurlabs) and industry surveys. Treat as order-of-magnitude.

- "OEKO-TEX added total fluorine (PFAS) limits in Standard 100 Edition 2024" — confirm exact PFOA/PFOS/total-fluorine values against the current OEKO-TEX Annex 6 before quoting in a regulated context.
- France's proposed €10/garment ultra-fast-fashion malus by 2030 is a Senate bill, **not yet enacted**.
- The CAP Act (raising CPSIA per-violation cap to \$250,000) was introduced 25 Jan 2024 and is **not enacted** as of mid-2026.
- SB 707 fee schedule and eco-modulation grid have **not yet been published**; specific design criteria reducing fees (recyclability, durability, mono-material, recycled content, PFAS absence) are inferred from statutory text and Anthesis's analysis.

Appendix: Threshold Reference Table (Engine C rule corpus, selected)

Reg.	Statute / Citation	Substance	Matrix / Scope	Threshold	Effective
CPSIA §101	15 U.S.C. § 1278a; 16 CFR §1500.87	Lead (total)	Children's product substrate	≤ 100 ppm	14 Aug 2009
CPSIA §101	16 CFR § 1303	Lead in paint/coating	Children's product surface coating	≤ 90 ppm (0.009%)	14 Aug 2009
CPSIA §108	16 CFR § 1307	DEHP, DBP, BBP, DINP, DIBP, DPENP, DHEXP, DCHP	Accessible component, toys/childcare	≤ 0.1% (1,000 ppm) each	25 Apr 2009
FFA	16 CFR 1610	Flame spread	Wearing apparel textile	Class 1 or 2 required; Class 3 prohibited (plain < 3.5 s; raised < 4.0 s with base burn)	Continuous
FFA	16 CFR 1615/1616	Char length	Children's sleepwear 0-6X / 7-14	Avg ≤ 7 in; max single ≤ 10 in; tested	Continuous

Reg.	Statute / Citation	Substance	Matrix / Scope	Threshold	Effective Date
				pre + 50-wash	
CA AB1817	H&S Code §108970	Regulated PFAS (Total Organic Fluorine)	Textile articles incl. apparel	≤ 100 ppm (2025); ≤ 50 ppm (2027)	1 Jan 2025 1 Jan 2027
NY S1322/A994	Env. Cons. Law §37-0121	Intentionally added PFAS	Apparel (excl. severe-wet outdoor)	Prohibited	1 Jan 2025
Maine LD 1537	38 MRSA § 1614	Intentionally added PFAS	Textile articles	Prohibited	1 Jan 2025
MN Amara's Law	Minn. Stat. § 116.943	Intentionally added PFAS	Juvenile products + textile furnishings	Prohibited (Phase I); reporting 15 Sept 2026	1 Jan 2025
CO HB22-1345	CRS § 25-15-602	Intentionally added PFAS	Textile articles + outdoor severe-wet	Disclosure (2025); ban (2028)	1 Jan 2025 2028
RI H 7356	RI law	Intentionally added PFAS	Textile articles	Prohibited	1 Jan 2025
VT S.25	VT law	Intentionally added PFAS	Apparel	Prohibited	1 July 2025
CT PA 24-59	CT DEEP Final Labeling Order (Dec 2025)	Intentionally added PFAS	Apparel + 11 categories	Labeling ("Contains PFAS" / "Made with PFAS Chemicals")	1 July 2025
CA Prop 65	H&S §25249.5; OEHHA	Lead	Daily dose	MADL 0.5 µg/day	Continuous
CA Prop 65	OEHHA	Cadmium	Daily dose oral	4.1 µg/day	Continuous
CA Prop 65	OEHHA	DEHP	Daily dose	NSRL 310 / MADL 410	Continuous

Reg.	Statute / Citation	Substance	Matrix / Scope	Threshold	Effectiv
				µg/day	
CA Prop 65	OEHHA	DINP	Daily dose	NSRL 146 µg/day	Continu
CA Prop 65	OEHHA	Formaldehyde	Daily inhalation	NSRL 40 µg/day	Continu
CA Prop 65	OEHHA	BPA	Daily dermal	MADL 3 µg/day	Continu
REACH §43	Reg (EC) 1907/2006 Annex XVII Entry 43, App. 8	Each of 22 carcinogenic arylamines	Textiles/leather, direct prolonged skin contact	≤ 30 mg/kg each	Continu
REACH §47	Annex XVII Entry 47	Chromium VI	Leather skin-contact	≤ 3 mg/kg	Continu
REACH §50	Annex XVII Entry 50	8 PAHs (BaP et al.)	Toys/childcare	≤ 0.5 mg/kg each	Continu
REACH §50	Annex XVII Entry 50	8 PAHs	Other skin-contact articles	≤ 1 mg/kg each	Continu
REACH §72	Reg (EU) 2018/1513	Formaldehyde	Textiles/clothing/footwear	≤ 75 mg/kg	1 Nov 20
REACH §72	Reg (EU) 2018/1513	Quinoline	Textiles	≤ 50 mg/kg	1 Nov 20
REACH §72	Reg (EU) 2018/1513	Disperse Blue 1, Orange 37/76/59, Yellow 3	Textiles	≤ 50 mg/kg each	1 Nov 20
REACH §72	Reg (EU) 2018/1513	Benzene	Textiles	≤ 5 mg/kg	1 Nov 20

Reg.	Statute / Citation	Substance	Matrix / Scope	Threshold	Effectiv
REACH §72	Reg (EU) 2018/1513	Cd, Pb, Cr(VI), As	Textiles	≤ 1 mg/kg each	1 Nov 20
REACH §72	Reg (EU) 2018/1513	DMF, DMAc, NMP	Textiles	≤ 3,000 mg/kg each	1 Nov 20
REACH §72	Reg (EU) 2018/1513	Σ(DEHP+DBP+BBP+DIBP)	Textiles	≤ 1,000 mg/kg	1 Nov 20
REACH §77	Reg (EU) 2023/1464	Formaldehyde emission	Furniture/wood articles	≤ 0.062 mg/m ³	6 Aug 20
REACH §77	Reg (EU) 2023/1464	Formaldehyde emission	Other articles (textiles exempt — Entry 72 binds)	≤ 0.080 mg/m ³	6 Aug 20
REACH §27	Annex XVII Entry 27	Nickel release	Skin-contact metal	≤ 0.5 µg/cm ² /week	Continu
OEKO-TEX 100 Class I	Edition 2025 (Annex 4)	Formaldehyde	Babies ≤ 36 mo	< 16 mg/kg	1 Apr 20
OEKO-TEX 100 Class I	Edition 2025	pH	Babies	4.0–7.5	1 Apr 20
OEKO-TEX 100 Class I	Edition 2025	Extractable Pb, As	Babies	0.2 mg/kg	1 Apr 20
OEKO-TEX 100 Class I	Edition 2025	Extractable Cd	Babies	0.1 mg/kg	1 Apr 20
OEKO-TEX 100 Class I	Edition 2025	Extractable Hg	Babies	0.02 mg/kg	1 Apr 20
OEKO-TEX 100 Class I	Edition 2025	Banned arylamines	Babies	≤ 20 mg/kg each	1 Apr 20
OEKO-TEX 100 Class I	Edition 2025	Phthalate sum	Babies	≤ 0.05% sum	1 Apr 20

Reg.	Statute / Citation	Substance	Matrix / Scope	Threshold	Effectiv
OEKO-TEX 100 Class I	Edition 2024+	PFOA / PFOS / total fluorine	Babies	PFOA ≤ 0.025 mg/kg; PFOS ≤ 1.0 µg/m²	1 Apr 20
GPSR	Reg (EU) 2023/988	General product safety	All non-food consumer products	Risk assessment + EU Resp. Person + Safety Gate notification	13 Dec 2024
ESPR	Reg (EU) 2024/1781	DPP	Textiles (per delegated act)	DPP required	~2028 (i adoptio ~2027)
CA SB 707	Pub. Res. Code §§ 42985-42985.27	Textile EPR	Apparel/textiles sold in CA	Per-unit eco-modulated fee; Landbell USA PRO (27 Feb 2026); registration 1 July 2026; enforcement by 1 July 2030; \$10K/day (\$50K intentional)	Phased